TUTORIAL 6 - Write a C++ program to overload the * operator to multiply two complex numbers.

NAME - Arenyu Bahiwal
BATCH - SY2 (A-2)
PRN - B25ET1073

```
#include <iostream>

using namespace std;

class Complex {
private:
    double real;
    double imag;

public:
    // Default and parameterized constructor
    Complex(double r = 0.0, double i = 0.0) : real(r), imag(i)
{}

    // Overloading the * operator using a member function
    Complex operator * (const Complex& other) {
        double r = (this->real * other.real) - (this->imag *
other.imag);
        double i = (this->real * other.imag) + (this->imag *
other.real);
        return Complex(r, i);
    }

    // Function to display the complex number cleanly
    void display() const {
        if (imag >= 0)
            cout << real << " + " << imag << "i" << endl;
        else
            cout << real << " - " << -imag << "i" << endl; //
Handles negative imaginary numbers nicely
    }
};

int main() {
    // Create two complex numbers: c1 = 3 + 4i and c2 = 1 + 2i
    Complex c1(3.0, 4.0);
    Complex c2(1.0, 2.0);

    cout << "First complex number: ";
    c1.display();

    cout << "Second complex number: ";
    c2.display();
```

```
    // Call the overloaded * operator
    Complex result = c1 * c2;

    cout << "Result of multiplication: ";
    result.display();

    return 0;
}
```

Output

```
First complex number: 3 + 4i
Second complex number: 1 + 2i
Result of multiplication: -5 + 10i
```